

நமது வனம் NAMATHU VANAM

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NAMATHU VANAM
AUG 2026 - OCT 2026
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**TAMILNADU ASSOCIATION OF SENIOR PROFESSIONALS
OF ENVIRONMENT AND FORESTS
(TASPEF)**



நமது வனம்

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Editor's Desk

Pallikaranai Marshland (PML) wetlands, a protected Ramsar Site, under the control of Conservation Authority of Pallikaranai Marshland, Tamil Nadu Forest Department, is in the news as the Brigade Group and Morgan Asia Pvt Ltd, builders were cleared for undertaking construction of a multi-story high rise building and later the permission to construction was withdrawn citing the reason that the proposed area of construction falls within the zone of influence of the PML.

The honour of being the first Ramsar sites declared in our Country goes to the Chilika Lake in Odisha and Keoladeo National Park in Rajasthan. Both were declared as Ramsar sites on October 1, 1981. The 100th Ramsar site, the Jai Prakash Narayan Bird Sanctuary (Surha Tal) in Ballia, Uttar Pradesh, joined the list on June 5th, 2026. Our Hon'ble Prime Minister Sri Narendra Modi famously declared on X, **"A century as far as Ramsar sites are concerned!"**

As of January 2026, there are 2,520 Ramsar sites around the world, protecting over 253,075,668 hectares (25,30,756.68 Sq. KMs). In India, we have a total of one hundred Ramsar wetlands covering an area of 1,387,612 hectares (13,876.12 km²), while, the State of Tamil Nadu has the highest number of Ramsar wetlands, a total of 20 sites, covering 1000.81 Square Kilo meters.

Any wetland, that meets any one of the 9 criteria needed for being considered Internationally important, can be declared a Ramsar site, under Ramsar Convention. Ramsar Convention, also known as "The Convention on Wetlands" an International Environmental Treaty, under the auspicious of UNESCO, was signed on 2nd February 1971, in a place by name, Ramsar in Iran. Hence the name Ramsar Convention. It came in to force on 21st December, 1975. It provides for national action and international cooperation regarding the conservation of wetlands and wise sustainable use of their resources.

Pallikaranai wetland is the only surviving freshwater ecosystem in South Chennai, located in the centre of a thriving commercial and residential localities like Sholinganallur, Thoraipakkam and Velachery. From an estimated extent of 5500 Ha., in 1965, the Pallikaranai wetland has now shrunk to about 698 Ha. This has been declared as Pallikaranai Marsh Reserve Forest to protect it from further loss of area. This area of 698 Ha of Reserve Forest and an additional area of about 550 Ha. Adjoining the marsh has been notified as a RAMSAR site no. 2841, as on 8th April 2022 with an area of approx. 1248 Ha. (Source: Web site of The Tamil Nadu State Wetland Authority) due its ecological importance. The press release No: 2569 Date: 28.10.2025 of Environment, Climate Change & Forest Department, Government of Tamil Nadu states that *"...the designated Ramsar Site includes 698 hectares of the existing Pallikaranai Marsh Reserve Forest and 550 hectares of additional area to be delineated after following due process, thus totalling 1248 hectares. The "ground truthing" as per the Wetland Rules for delineating boundaries with specific survey numbers of land parcels other than Pallikaranai Marsh Reserve Forest have not been done so far..."*

The reason that Pallikaranai wetlands is of late in the news, is due to the objections raised by a NGO and a Public Interest Litigation filed in the Honourable Madras High Court, objecting to the

construction permit granted for a multi-story building within the one Kilometre zone of influence. The southern bench of the National Green Tribunal (NGT) *Suo Moto* initiated the complaint and passed an order, banning any developmental activities within the zone of influence.

The Confederation of Real Estate Developers' Associations of India (CREDAI), has challenged the above construction ban claiming the fixation of zone of influence was "without" any scientific survey or preparation and approval of the Integrated Management Plan as mandated under the Wetlands (Conservation and Management) Rules, 2017. The Brigade Enterprises Limited and Another, have preferred a Writ Petition before the Honourable Supreme Court of India, challenging the revocation of the Environmental Clearance (EC) granted to it for undertaking their building project. The Honourable Supreme Court of India vide its order dated 17.07.2026, dismissed the Writ Petition by declining to grant any relief (Writ Petition (Civil) No.749/2026).

Conservation, development, livelihood and property right issues all are intermixed in this delineation of one-Kilometre-wide zone of influence all around the Pallikarani Wetlands. All stakeholders are keenly awaiting the decisions of the honourable courts.

As of now, it is learnt that the Tamil Nadu Forest Dept has not notified a formal Eco Sensitive Zone (ESZ) under Environment Protection Act 1986 for Pallikarani. They are managing it under Pallikarani Marshland Reserve Forest rules. We are sure, the State Wetland Authority would do all that is needed, while finalising the boundaries of Zone of Influence, taking in to consideration all legal and social implications that has arisen due to freezing of all planning permissions by Chennai Metropolitan Development Authority (CMDA).

Wetlands are important and critical ecosystem which hosts unique biodiversity and are a haven for avian fauna and other aquatic & amphibian life forms. Declaring a Wetland as a Ramsar site is to promote the conservation of Wetland Ecosystem giving an impetus to conservation strategies. Unfortunately, several wetlands are close to habitations and industrial belts. We may recall that few years back, Vedanthangal Bird Sanctuary was in the news about the expansion plans of Sun Pharma company which was located well within 5 Km radius of notified Sanctuary.

It is quite necessary that a wide consultation process takes place between business and residents surrounding the zone of influence and a ground truth-based zonation is defined and settled with the consent of all the stake holders. Otherwise, prolonged litigations will affect the efforts of conservation of Wetlands and may face negative attitude from the local public which is not conducive to the biodiversity conservation. Thankfully as the PML issue is now legal issue and *sub judice under judicial process*, we only wish to flag the importance of zonation and delineation of boundaries is a very essential and critical part of any Management Plan of any Wetland or Sanctuary or National Park. A consultative process involving all stakeholders in fixation of boundary should be the purpose of our Conservation Plans.

TASPEF hopes that the concerned authorities will learn from the lessons of past and promote congenial Conservation Strategies.



V. Prabhakaran, IFS., (R)
Former Additional Principal
Chief Conservator of Forests

TASPEF Greets incoming and outgoing HoFF, PCCF, Tamil Nadu



The present Principal Chief Conservator of Forest and HoFF Thiru Srinivas R. Reddy, IFS., retires on 31.7.2026 on superannuation. He belongs to 1991 TN IFS batch. Mr.Reddy held the post of HoD of TNFD from 1.10.2024 with distinction initiating several development initiatives in the Forest Department. TASPEF warmly greets Thiru Srinivas R. Reddy, IFS., on successful completion of his service and wishes him a cheerful retirement life.



Thiru A. Udhayan, IFS., a 1991-batch Indian Forest Service (IFS) officer of the Tamil Nadu cadre has been appointed as the Principal Chief Conservator of Forests and Head of Forest Force (HoFF) for Tamil Nadu. He will be taking official charge on August 1, 2026, following the retirement of Thiru Srinivas R. Reddy. He is presently Director, Advanced Institute for Wildlife Conservation Vandalur since June 2022. Previously he was Chairperson of Tamil Nadu Pollution Control Board (TNPCB) from October 2021 to June 2022. Thiru A. Udhayan also rendered his services in SAARC Forestry Centre in Bhutan. Thiru A. Udhayan holds an M.Sc Degree in Environmental Sciences from JNU, a PG Diploma in Wildlife Management from WII, and completed a Public Policy and Management program at IIM Bangalore. His expertise spans in the field of wildlife management, biodiversity conservation, public policy, and environmental governance

TASPEF extends warm greetings on his appointment as Head of Tamil Nadu Forest Department. He is also Ex-officio Patron of our Association as per our Bye-law .We wish him all the best in his new responsibility and we are sure he will give the right direction and leadership to the Forest Department.

We in TASPEF are eager and happy to cooperate with him and assure to place our services in whatever manner required. TASPEF heartily welcomes our new Patron.

REMINISCENCES OF A FORESTER – SOME SNIPPETS (Episode 6)

It was general transfer time in 1993. Normally officers who have completed their tenure in a post, which was usually then three years, would anticipate a shift from their current station. In my case, I was just completing two years in the post of DFO, Tirunelveli division. Still, I expected that axe would fall on my official position for the reasons explained in the previous episode. This time around I couldn't shift my family, as my wife needed to avoid travel for health reasons. By end April 1993, order for my transfer to Tiruchirappalli Working Plan division was received. My aged mother couldn't be with my wife and my five-year-old daughter at that crucial situation to take care of them. My in-laws had unfulfilled family responsibilities including marrying off their youngest daughter. Per force I had to enter on leave to take care of my wife. That was probably the first and only time I took long leave of about four months in my career.

By August 1993, I received a Government Order posting me as Divisional Forest Officer, Kamaraj Social Forest division headquartered at Virudhunagar. It was for the first time; I got positioned outside the realm of territorial forest management. The division was named after Late Kamaraj, the freedom fighter, veteran Congress leader and Chief Minister of Tamil Nadu, whose native town is Virudhunagar. The town, an important trading center in the southern Tamil Nadu, was earlier known as Virudhupatti, which got christened to its present name in 1923 in the town council meeting. Though strategically

located on the longest National Highway (previously NH 7), connecting Kashmir with Kanyakumari and despite being the district headquarters, the town didn't outsmart any of its counterparts in terms of urban infrastructure, as if to honour its original name of 'patti' as surname (literally meaning hamlet). Kamarajar district got trifurcated in 1985 from the erstwhile Ramanathapuram district, known to be the driest tract of the state.

I moved to Virudhunagar soon after joining the division, where I got accommodation in the TN housing board quarters. The Divisional Forest Office was functioning from a private building close to the Virudhunagar Senthikumarar Nadar College (VHNS), about 5 km from the premises of the District Collectorate, where the quarters were located. Major works in the division included raising tank foreshore plantations in the panchayat and PWD tanks and some tamarind plantations in other panchayat lands. Another primary function was to assess the yield of babul plantations and sell the same through public auctions. From the stress bound work, typical of a territorial division, I thought that the responsibility of social forestry DFO was less challenging. That gave me free time to think about unconventional job assignments.

Study of Nandavanams (Community Groves)

Virudhunagar district boasted of many sacred groves within the limits of most of the taluk towns, which frequency I didn't come across in any other district

where I served earlier. These groves, which spanned to the extent of few cents to many acres supported an assortment of natural and planted trees other than the intermittent floral garden within the grove. Invariably some local deities are identified with each grove, hence it is sacred to the community that established the same and maintained with reverence. In addition to religious attachment, the community also maintained common infrastructure such as bathing ghat with open well and water lifts (manual those days) open to the public of the town.

I was inquisitive to understand this unique practice in the district and commissioned a survey of community Nanda Vanam in Virudhunagar district, by engaging a masters in Botany from the local college. When completed, the study contained an account of over 40 groves spread over six towns of the district. Thus, my tenure in a social forestry division gave me an opportunity to prepare an insightful document on one of the cultural heritage traditions of the region. It is another matter that with steep rise in land value in urban areas, the Nanda Vanams of the towns gave way to kalyana mandapams and community halls, leaving no trace of the rich eco-cultural past.

First encounter with the construction of official residence for the DFO

One of the immediate things that sought my attention, when I joined the division was to continue with the construction of the official residence of the Divisional Forest Officer on the site allotted for DFO quarters within the Collectorate Master Plan Complex. Thanks to the effort of the then DFO, SF division Mr. Samant Singhar IFS., 75 cent plot was allocated to the forest department for putting up the

quarters. As the land was readily available, department allotted about Rs 4.0 lakh for the construction during 1993-94. It was a humongous task to carry on with the construction of about 1800 sq.ft. built up area with G+1 design at that estimated cost (worked out at Rs 220 per sq ft). Only comforting fact was that my predecessor Mr. N. Krishna Pillai, IFS., picked up a reasonably good person from among the Babul plantation contractors.

Since my TNHB quarters was within walking distance of the upcoming project, I was able to supervise the work closely. Even with minor overshooting of timeline, I was able to get the building completed in a year's time since my joining. Named *Kanagam*, we had the housewarming, graced by the Tirunelveli Social Forestry Circle Conservator of Forests Mr. K.S. Dehal, IFS., in the second half of 1994 and soon thereafter I moved in with my family. My driver Murugan was an avid gardener, and he was meticulously assembling a decent floral and vegetable garden even when the building was rising. The premises began to gain attraction from the nearby community of assortment of Government officials with their families. Later when I visited the campus after nearly two decades, I noticed that the residential building was still in its prime status with minor maintenance interventions.

Muddle-Puddle planting

Virudhunagar district is basically a semi-arid tract and receives average annual rainfall that is often less than state average. During my tenure in the division from 1993 to 1996, the year 1995 turned out to be particularly deficient year with scanty rain. Though the district receives rain from both the monsoons, of the total volume north-east monsoon records 53 per

cent of the year total. I had a target of 550 ha of tank foreshore plantation for 1995-96, spread in the six ranges of the division. Taking advantage of the sporadic rains, Forest Range Officers of the division were able to somehow complete the planting works in most tanks by November.

There was rainfall aberration in a couple of villages in Sivakasi range, where the targeted tanks didn't receive adequate inflow to support the planting. The Ranger Mr Nandagopal was waiting with the nursery stock that was overgrowing day by day. Half size pits were ready in a 40-ha plantation site but not sufficient to accommodate the plant. No sign of rain too. The Range Officer came up with an idea of creating a muddy-puddle with one leg in each of the designated pits by pouring water, sourced from a nearby well through tractor-fitted tanker and plant the babul seedlings from the 16x30 cm poly bags. He shared the thought with me, and I pondered as to why we couldn't attempt. I asked him to proceed.

It so happened that a circle review meeting of the DFOs by the Chief Conservator of Forests (CCF) of Social Forestry wing Thiru P.K. Ponnusamy, IFS at Tirunelveli coincided with this time. I presented the concept of the muddy-puddle planting of babul in my division before my colleagues and seniors in the meeting, which drew sarcasm and was received with a pessimistic note. The senior officials were unwilling to buy the concept of **muddy-puddle planting**, unheard of so far and blessed me with success for this unconventional wisdom. The science behind the creation of the puddle over a compact space of say 30 cm diameter is that the precious water droplets, entrapped to the bottom of the puddle, will remain for a reasonable time to retain moisture around

root region and provide succor to the transplanted seedling. This contrasts with a larger muddy puddle where the broader surface area tends to evaporate water particles faster when in contact with heat and air. At the end of the financial year, when the staff enumerated survival in the said plantation, we were happy to notice only normal number of casualties. Plants too established reasonably well. Thus, the experiment proved to be an eye-opener to change the threat into an opportunity.

Narkaruvel vs Murkaruvel

Babul (*Acacia nilotica*) used to be referred to as narkaruvel to mean that this species is devoid of the prickly thorns that are typical of seemai karuvel (*Prosopis juliflora*). However, one variety/sub-species of babul with distinctly sharp and bold thorns on the main stem and branches happened to be prevalent along with less-spiny narkaruvel in many tank foreshore plantations in the saline-alkaline patches of the tanks in Aruppukottai, Thiruchuli SF Ranges and to some extent in Sivakasi SF Range. This variety/sub-species is deterred by the babul felling contractors because of the sheer difficulty in working the plantation and therefore, usually the last batch of plantations to be claimed by the bidding contractors in the open auction.

Having seen the inherent disadvantage with this sub-species, I began pestering my Range Officers to source *Acacia nilotica* seeds from pure stand of narkaruvel for all fresh plantations, result of which was seen distinctly in the nursery stage itself.

Our tryst with wedge prism

In an earlier episode, I narrated my work experience in assisting the Conservator of Forests for the preparation of volume table

for *Eucalyptus globulus* in the Nilgiris circle. It was the same Conservator Thiru Rajagopal Shetty who after extensive painstaking field work with babul had given the yield table for the species, a few years back (*Acacia nilotica*). As Babul was a good revenue source for the department, all social forestry divisions get into a flurry of activities around early January each year. With reference to the rotation cycle of eight years that was in vogue then, the DFO would list the babul plantations due for harvest during the ensuing financial year, get the plantation assessed for its fuel wood yield per ha, prepare a fair price for the plantation taking into account the market price, cost of felling, conversion and transport and of course the contractor's profit, get it approved by the competent authority (Conservator of Forests/Chief Conservator of Forests, Social Forestry, as the case may be), publish the sale notice in the district official gazetteer and conduct the sale proceedings.

The yield assessment followed a rigorous scientific method of demarcating the boundary line with reference to the original survey sketch of the plantation, laying base line, grid line within the plantation as per sampling norms so that we get x number of sample points per ha, depending on the extent of plantation. This is followed by use of wedge prism at the sample point to determine the number of tallied trees and of altimeter to record the height of average sized tree. All the data recorded thus are read with the volume table that would give the volume of wood obtainable per ha at that point. Volume gets converted to weight using the standard conversion formula and anticipated yield in ton/ha is computed and fair price/ha arrived.

To ensure proper countercheck at each

administrative level, DFOs and CFs were required to randomly check the fair price. DFO needs to check the same in 50 % of the plantations under his control and the CF 10 %. This used to be a time-bound operation, as any delay could make the tank unavailable for check because of rain that submerged the plantation. The task occupied the priority task list of the DFO and CF in the summer months in social forestry wing.

The CF Mr. K.S. Dehal, IFS undertook the super check in a rigorous manner so that no plantation of higher outturn is left unchecked by him to prevent any under assessment and valuation. As we say, that 'proof of pudding is in its eating', the nearness of field estimation to actual output of fuelwood will get confirmed when the fair price and sale price match each other, except for +/- 5 to 10 percent variation. If the unit had fetched unexpectedly higher auction price, that showed that something went amiss in the field assessment, exception being unhealthy competition among bidding contractors. When across the State such high anomaly occurred, it proved invariably that something went wrong in the field work of the subordinates mostly because of wrong laying of grid lines and fixing of sample points either by mistake or by design. Intentional underestimation always led to stiff disciplinary action against the individual official.

- To be continued



Dr. T. Sekar, IFS., (R)

Former Principal Chief Conservator
of Forests

TIGERS UNDER THE BRIDGE



The monsoonal winds and the buildup of clouds started to cool the surrounds and on the 5th of June 2013 it had decidedly become cooler, simmering down the fierce heat and settling the choking dust of May. In its wake came the saturation of moisture and it was humid. A few days ago we bemoaned the heat and dust, and now we were stifled by the sudden change in the weather; from the dry desiccating heat to smothering humidity. It was the day celebrating the World Environment Day, and we were stirring up to the new born day.

The sun was now peeking from the gaps between the slowly drifting dark, low clouds. In this mood of the day, news broke that there was a tiger under a culvert being constructed. The first reaction was for the safety of the labour and people using the road. Bhramapuri, from where the news came is famous for tigers for all the wrong reasons. The news threw us all into high gear. Safety was the first priority – for the people and for the tiger!

The day I joined the Chandrapur Circle as its Chief Conservator of Forests (CCF) on the 7th of June I received the news

and it became my priority to see that no untoward incident took place, especially since it happened during the cusp of handing over and taking over charge of an area fraught with wild life conflicts – especially with leopards and tigers – tigers mainly. The area was where the jurisdictions of the Forest Development Corporation of Maharashtra (FDCM) and the Territorial Forests of Brahmapuri meet, the canal being the boundary. My first choice was Kamlesh Thakur, a young, energetic wild life photographer; driven, passionate and committed.

Immediately Kamlesh was intimidated about the task at hand for him. He was required to scout the area, check for movement of the tiger and report back every detail. He is enthusiastic and knows the lie of the land and can use it to his advantage. Being well equipped, he was the best choice at that point to record the goings on. Then news came in that there were cubs under the culvert. That was a situation no one wanted. Cubs – tigress – people is a very bad mix even in the best of times. Here with a culvert under construction would jeopardize the fragile balance of conflicts. Nothing could be predicted, hence we took every precaution and let the events unfold, and we would take informed decisions based on local knowledge and what Kamlesh would report back, possibly with photographic proof. The canal in question was the Gosikhurd Canal. The area under the FDCM falls in the Pathri Range.



While Kamlesh set about his task, I ordered the setting up several camera traps to record the goings-on. We also requisitioned the services of Uday Patel who was the former Honorary Wildlife Warden of Chandrapur, lately of Gadchiroli, a man who knows the forest like the back of his hand, having walked in the forests almost his entire adult life. In fact, there might not be a living person today who has as much intimate knowledge of the terrain and also the animals – including leopards and tigers – of that pocket of Central India.

In the meantime, Zade, Range Forest Officer (R.F.O.) of Sindewahi (FDCM) and other field officers' came to the spot. There was a crowd building up, but the tiger remained calm, much to the surprise of everyone. It later turned out to be the tigress and mother of the four cubs.

Kamlesh is a very plucky man; there were pugmarks of a tiger. Kamlesh peeped into the tunnel and when his eyes adjusted to the gloom, he saw a tiger! Barely twenty feet from him and seemed quite unperturbed. All this when it was not known that the cubs were there!

That evening it rained. The culvert would have been an ideal place for the cubs since the entrance was partly walled by the mud, but given the circumstances, it was not conducive and potentially very dangerous!

Observers posted nearby noted that they heard the calls of cubs coming from under the culvert. The next day the tigress was not present so a few of us led by Kamlesh crawled one after another into the tunnel and saw four tiny tiger cubs whose eyes had just about opened. Since the rains would flood the canal, the cubs were in danger of drowning since the floor of the culvert was becoming slushy with water.

With heavy rains predicted, we made haste and cleared the canal as best we could. I had just taken over charge and I did not want any untoward incident to mar my joining as CCF Chandrapur.

The weatherman had predicted heavy thundershowers, the clouds threatened to burst any moment, so there was scarcely any time to think out a strategy. All hands onboard, elbow grease and rolled up

sleeves was the need of the hour. Shoulder to shoulder, Uday, Abhishek Deshmukh and I set to work. I ordered the forest staff under me to build a mound at the entrance of the canal under the Gosikurd box canal culvert and we built a platform so that the tigress could get in and out while the mound kept the water out. A diesel pump was got and placed there to suck up the water if the canal filled up ever so slightly.

Barely had we finished when we heard the tigress call. We hurried away. We did not leave the place, but moved away and were watching from a distance when camera traps started to flash. The mother sat on the upper side, waited there. From the behind our jeep came the male; the father.

His presence seemed to have instilled confidence in her. He went in to the adjacent concrete water course. She followed. We could only see the camera flashes going off in rapid succession. She brought the cubs out one by one and gathered all the four under a bamboo clump.

Within the next half hour or so she moved all the cubs one by one beyond our prying eyes. Next day all our efforts to locate the family went in vain; the rains washing whatever pug marks were left on the forest floor. When cubs disappeared to safety designed by their mother they were barely few weeks old. The climax was that it rained heavily and the whole culvert was filled with water and the diesel pump would have been useless in the torrential rain and subsequent flooding, we in fact recorded twenty feet of swirling water that submerged the culvert and the road. The flood water would have drowned all the four cubs.



Our prompt action and the disturbance we created prompted the tigress to remove her cubs to safety and they were saved. The forest department monitored their movements for many months thereafter. The mother was still with them. The mother and cubs range over a large area. The male still roars in the same region but he has gone back to his bachelor days.

Providence played a major role because prior to taking over charge as CCF Chandrapur, I was General Manager of the area under the FDCM. My handing over office, and taking over office, the boundary a canal, a tigress, her four cubs and a tiger made my transition from FDCM to Territorial Forest very exciting and it augured in one of my best postings.



Mr. Sanjay P. Thakre, IFS., (R)

He is a retired Chief Conservator of Forests from Maharashtra. He has worked in interesting areas like sironcha, Gadchiroli, and Chandrapur. He was Principal of the Rangers' College, Chandrapur and later also the Field Director of The Tadoba-Andhari Tiger Reserve (TATR). He was also General Manager Forest Development Corporation of Maharashtra (FDCM) and member Secretary Maharashtra State Zoo Authority from September 2016 to June 2017 till he superannuated.

RUDRA'S CANOPY- A TALE OF TWO MOUNTAINS

*In the quiet depths of our ancient forests, there exists a tree that bridges the gap between the terrestrial and the divine. To the botanist, it is *Elaeocarpus*; to the seeker, it is **Rudraksha**—the **Eye of Rudra** or the **Tears of Shiva**. Legend tells us that after a thousand years of meditation for the welfare of all beings, Lord Shiva opened his eyes, and the tears that fell to the earth blossomed into these sacred trees. Today, as we look at the misty peaks of the Himalayas and the verdant ridges of the Western Ghats, we find that these “tears” are in need of our protection.*

Sacred Seed

India is home to a rich diversity of *Elaeocarpus* species spread across two distinct biological hotspots. The genus name itself, derived from the Greek *elaia* (olive) and *karpus* (fruit), hints at the stony, olive-like fruit that characterizes the family. India is home to approximately **25 to 35 species** of *Elaeocarpus*.

Feature	The Himalayas & Northeast India	South India (Western Ghats)
Species Richness	High (Approx. 18–26 species)	Moderate (Approx. 9–12 species)
Endemism Level	Lower (Many widespread species)	Very High (Approx. 80% endemism)
Key Habitats	Sikkim, Arunachal Pradesh, Assam, and Meghalaya	Tropical evergreen and montane forests (e.g., Palani Hills)
Notable Species	<i>E. ganitrus</i> (Rudraksha), <i>E. lanceifolius</i>	<i>E. tuberculatus</i> , <i>E. munronii</i> , <i>E. pulneyensis</i>
Distinctive Traits	Includes species found across Southeast Asian ranges.	Characterized by massive buttressed trees (e.g., <i>E. tuberculatus</i>).
Conservation Status	Habitat loss in the foothills is a major threat.	<i>E. blascoi</i> Vulnerable due to restricted range and high endemism.

Endangered & Endemic Status

The Western Ghats house some of the most threatened members of this genus due to their narrow geographical range:

Critically Endangered- *Elaeocarpus gaussenii* (found in the Megamalai hills) and *E. blascoi* (Palni hills).

Endangered/Endemic- *E. venustus* is a rare endemic restricted to the swampy high-altitude forests of the Agasthyamalai hills (South India).

Threats- These trees face high risks from habitat fragmentation, climate change, and poor natural regeneration (rigid seed coats).

The Himalayan Sentinel- *Elaeocarpus ganitrus*

The most famous species, *Elaeocarpus ganitrus* (syn. *E. angustifolius*), thrives in the humid, subtropical climate of the Himalayan belt—spanning from Uttarakhand through Nepal to Northeast India. These trees can reach staggering heights of 60 to 80 feet, standing as silent sentinels of the mountains.

The fruits, which turn a brilliant “blueberry” blue when ripe, protect the seed that becomes the prayer bead. In the middle Himalayas, these trees are more than just spiritual symbols; they act as vital **nutrient filters**, enhancing the quality of water that seeps into the ground to feed the Ganga and its tributaries. Their extensive root systems stabilize the fragile mountain soil, preventing erosion during the fierce monsoon seasons.

The Western Ghats Endemics: *Elaeocarpus munronii* and *E. blascoi*

Further south, the ancient **Shola forests** of the Western Ghats harbor their own treasures. Here, species like *Elaeocarpus munronii* (often called the “Mock Olive”) and the critically endangered *Elaeocarpus blascoi* (Palani Hill Rudraksha) struggle for survival.

E. blascoi is a poignant example of the fragility of our biodiversity. It was once thought to be extinct until a few specimens were rediscovered near Kodaikanal. Unlike their northern cousins, these species are often “**steno-endemic**,” meaning they are found in extremely restricted geographic pockets—sometimes limited to a single valley or stream bank.

The Crisis in the Canopy

Despite their sacred status, Rudraksha trees face significant ecological threats that transcend regional boundaries. The “tears” of the forest are drying up due to several factors:

Habitat Fragmentation-

In the Himalayas, the replacement of native broad-leaved forests with commercially viable Pine has pushed Rudraksha to the brink. In the South, the expansion of tea and coffee plantations has fragmented the once-continuous Shola-grassland mosaic.

Anthropogenic Pressure-

Unregulated grazing, forest fires, and the over-harvesting of fruits for the bead trade have prevented wild patches from regenerating. When every fruit is collected for its market value, no seeds are left to become the next generation of trees.

Low Germination Rates-

The seeds have a naturally tough “stone” (the endocarp) and specific requirements for moisture and soil. In the wild, germination can take anywhere from six months to two years, and the survival rate of seedlings is notoriously low.

Climate Instability-

In the Western Ghats, changing rainfall patterns and increased soil erosion often wash away the few seeds produced by the rare *E. blascoi* before they can take root in the damp, mossy soil they require.

Cultural and Spiritual Significance

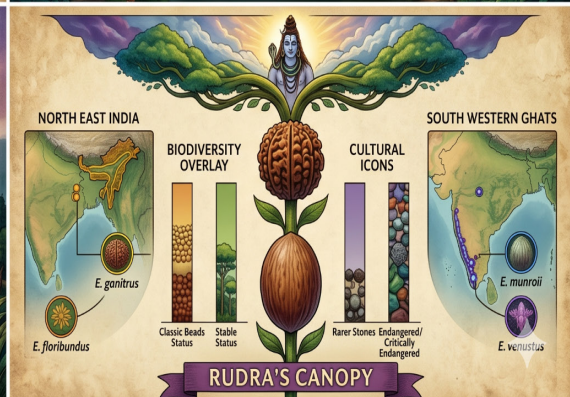
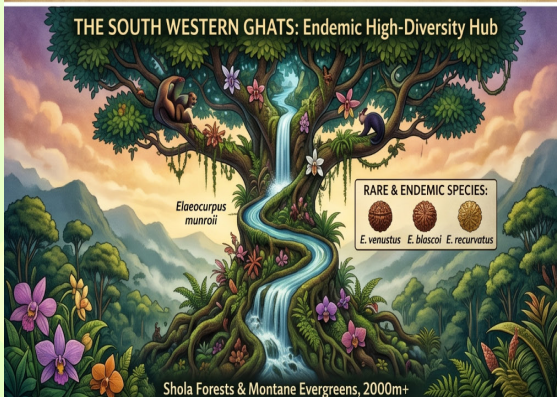
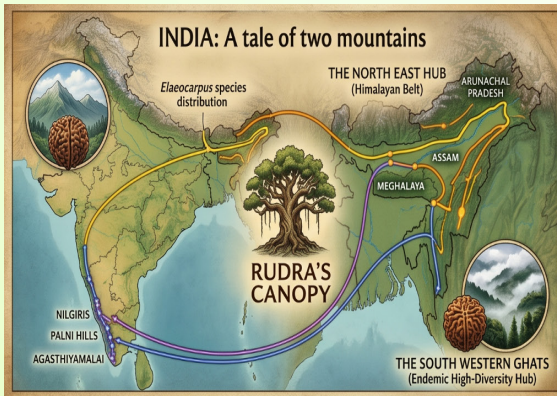
The value of the Rudraksha extends far beyond its botany. Culturally, it is the thread that binds the diverse tapestry of Indian spirituality.

- **The Beads of Consciousness-** The endocarp of the fruit is divided into segments called *Mukhis* (faces). In tradition, each *Mukhi* is believed to resonate with a specific planetary energy or deity, serving as a tool for grounding and focus during meditation.
- **Traditional Medicine-** Beyond the beads, the leaves and bark of *Elaeocarpus* species have been used for centuries in Ayurveda and Siddha medicine. They are recognized for their anti-inflammatory and anti-hypertensive properties, often referred to as the “King of Herbal Medicines” for their ability to calm the nervous system.
- **Ecological Stewardship as Worship-** Historically, many Rudraksha groves were protected as **Sacred Groves** (*Devrai* or *Kavu*). In these areas, cutting a tree was considered an act of sacrilege, ensuring that biodiversity was preserved through the lens of faith.

A Path to Conservation

Conservation is not just a botanical task; it is a restoration of our connection to the earth. To ensure these species thrive, a multi-pronged approach is necessary:

1. **Scientific Intervention-** Utilizing tissue culture and advanced nursery techniques to raise saplings of endangered species like *E. blascoi*. By mimicking the complex rainforest ecosystem—providing the right fungal associations and moisture levels—we can improve the survival rates of young trees.
2. **Support Native Reforestation-** Landowners and forest departments must prioritize planting indigenous *Elaeocarpus* species over ornamental or commercial exotics. A single native Rudraksha tree supports a vast network of local insects, birds, and soil microbes that a Pine or Eucalyptus tree cannot.
3. **Community Stewardship-** Engaging local communities in the Himalayas and the Ghats is the only sustainable way forward. By creating “community conservation areas,” locals can be trained to harvest fruits sustainably, ensuring enough seeds remain on the forest floor to sprout.
4. **Mindful Consumption-** As consumers, we must understand that every Rudraksha bead represents a tree that cooled the earth and purified the air. Supporting ethically sourced beads from cultivated groves, rather than wild-harvested ones from endangered patches, makes us part of the solution.



Conclusion

As we hold a Rudraksha bead in our hand, let us remember it is more than a tool for prayer. It is a living testament to the resilience of nature and a symbol of humanity's ability to restore what we have endangered. The survival of these species—from the heights of the North to the ancient ridges of the South—is a victory for our collective heritage. By protecting the “Tears of Shiva,” we ensure that the canopy of our future remains green, sacred, and whole.



Elaeocarpus ganitrus



E. E. variabilis



E. tuberculatus



E. serratus



Only One Standing on the Brink- The unique, wild Elaeocarpus blascoi tree at Pambar Shola, Kodaikanal—one of the rarest plant species on Earth, strictly endemic to the Palani Hills.



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MY EXPERIMENT WITH SINCERITY

Background:

We Officers with more than thirty years of service, irrespective of their cadre, might have faced challenges in our career while performing our duties sincerely which might be persisting in our minds as Green/Dark memory. Some of our sincere actions in our service might have hidden as untold stories or unrecognized. This may be due to the existing political system, conflicting / un-supporting nature of the superior officers during those periods. Bringing out such untold stories may bring out the real value of the sincerity of the officer and also expose the official system existing during those periods. The objective of this story is such incidents should not die with the officers as unexposed. In this respect one of our senior Officers, Mr. L. Nathan, IFS., has written a book which really exposed some of his dare and sincere actions during his service. Not only Mr. Nathan, every one of us might have faced challenges while performing their duties with sincerity. The Editorial chief and the team should encourage our retired officers to come forward to come with such stories which are tingling in their minds as green / dark memories.

I start the first section of my story on this the topic, 'My Experiment with sincerity'

My Experiment with Sincerity – Part- I

“The arrow is on the string, but the bow is not drawn.”

Here is my first story on this topic:

It was my first posting as a DFO / Wildlife Warden of the Srivilliputhur Wild Life Sanctuary. I served there nearly four years from 1991 -94. As any other young officer, I was very agile and active, walking and climbing the hills of the present Srivilliputhur part of Megamalai Tiger Reserve. Those times, even now, many parts of Srivilliputhur sanctuary are not accessible by roads. One has to climb several hours to reach the peak. There is only one a century old Rest House called, 'Mudaliar Oothu Forest Rest House (MUFRH)' at the top of the hill where one could get shelter. The area belongs to Srivilliputhur Forest Range.

Due to its remoteness, it was vulnerable for ganja cultivation and poaching those times. On taking charge of the division, I took up a challenge of eradicating ganja cultivation and made frequent visits to the top. The MUFRH is the only halting place from where one could perambulate the top forests. Whenever I visited with my team, there was clear case of ganja cultivation offence. Eradication of ganja cultivation incidents by us appeared in local newspapers frequently. The local police seldom took interest to involve in ganja eradication operations there as, it involved tedious journey by walk and required a lot of preparations.

On seeing the frequent ganja eradication news, the district level police officers might have felt guilty for their inaction. During that period one young direct IPS SP made a secret raid to the Ganja areas with his police battalion mainly consisting of Armed Reserve Police. With great difficulty, the police team headed by the SP reached the top by walk and stayed in MUFRH overnight. Their official purpose

was ganja eradication, and getting rewards for such operation. But the team could not find any ganja cultivation. However, they are very successful in hunting a Nilgiri Tahr, the schedule I animal. The team prepared a good meal with the Tahr meat that night and stayed overnight there. As a sign of their successful operation, they kept the removed skin of the animal under sun to dry ignoring or ignorant of, the serious legal implications of their action.

Incidentally during that period, I was conducting wildlife census with the help of Mayuram Wild Life Biology Students (WLB) team under the head of the WLB Professor. I asked the team to go over to MUFRH area to do census of the Nilgiri Tahr and other wild animals. On reaching the MUFRH, the census team, found the Rest house was occupied by the police team and also saw the freshly hunted whole skin of the tahr. The police team under the head of the SP was caught red-handed by the census team. But the census team could not do anything as the SP was heading the team.

After the census, the Professor's team came down with the seized fresh Nilgiri Tahr skin and reported to me. They explained the entire episode. I was very serious over the poaching incident and I asked the Srivilliputhur RO to register the hunting case with the seized skin of the Tahr and got the statement of the Professor. As the news of booking of wildlife hunting case against the police team spread to the local police and the SP, the local police started showing enmity with the Forest Department. When I discussed the matter with the then Conservator of Forests at Tirunelveli, on how to go about, he advised me to keep the case in low profile as it involved the SP.

The SP, after knowing the seriousness of the case which carries imprisonment if convicted, planned to book a false brothel case against me, if I proceeded to prosecute the case against him and the other involved police personnel. This matter reached me through another forest officer. Although, I was keen in pursuing the case, since I was about to complete my tenure and my local staffs were concerned about the continuing enmity attitude of the local police with local Forest Department, I kept the case in low profile but live.

After a few years of my relief from post of Wildlife Warden, Srivilliputhur, I asked the concerned Ranger the status of the case. The reply was that the case was treated as Undetected Offence Report (UDOR). Sometimes the Forest Department faces very peculiar vulnerable and tricky circumstances in discharge of duty.



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THE TOUGH OFFICER WHO ENJOYED THE WONDERS OF NATURE.

Sometime during 1981, we were informed that the CF, Mr. Sivasailam, the tough Officer wanted to stay at Sengaltheri, the hill top on the Kalakad hills with an altitude of 940 m MSL. The good old rest house also known as 'Singalam Illam' probably named after the 'Lion-tailed Macaque' built there during 1906-07 is sandwiched between the grasslands and undulating hills in its front and dense forest on its back in the core zone. The cost of construction is mentioned as Rs. 2687 on one corner of the front wall. The Bungalow Watcher and the beat staff were informed to keep the rest house clean for the CF's stay. The Forester also was instructed to be ready to attend the camp.

The particular day also came and the CF reached Sivanaperi (about 2 kms from Kalakad) where the Forest Range Office, Range Officer's Bungalow and other staff quarters were located around 4.30 pm along with the WLW, Mr Thirumurthy. I received them and accompanied to Sengaltheri as the other field staff had already gone over there. By the time we reached the rest house, it was about 5.30 pm. After relaxing for a while, they occupied the cane sofas put up in the veranda and started enjoying the refreshment and the scenic beauty of the vast stretch of undulating hills covered with grass interspersed with younger regeneration of Phoenix and a few stunted trees including a solitary Rhododendron.

The map of Kalakad RF was shown to the CF as he evinced some interest in map reading. I gave the books 'Indian Animals' by S.H. Prater, 'The Book of Indian Birds' by Salim Ali, 'Tiger' by Kailash Sankhala and 'The Deer & the Tiger' by George B. Schaller, considered to be the popular books on wildlife, by wildlife enthusiasts

to the CF just for reading them casually. I said that Kailash Sankhala was the son of a Forest Range Officer of Rajasthan state. Of course, they were my personal books. Then, after dinner the CF and the WLW occupied the suites allotted to them.

Next day morning, after breakfast, it was planned to take the CF and the WLW to Karumandi Amman temple, a small temple meant for the Goddess on the bank of Karumandi Amman River, a perennial one that flows about a km away on the south-western side of the rest house. Immediately after leaving the rest house, at the rear side of the rest house, we could notice fresh dung of elephants that had visited the nearby water course that flows behind, the previous night. During the upward trek, we could watch a troop of Nilgiri langurs in the dense forest at the backside of the rest house. They jumped from one tree to another, travelled mostly on the tree tops and disappeared in a short while.

A solitary Indian Giant Squirrel was found moving on the top branch of a Wild Nutmeg or Kattu Jathikkai tree (*Myristica malabarica*). The moment it heard our voice, it hid its body on the upper portion of the branch and after some time jumped on to the nearby Nangu tree (*Mesua ferrea*), and disappeared. The Wild Nutmeg tree, which is endemic to the Western Ghats, is classified as 'Vulnerable' according to the IUCN Red list. The slightly aromatic arils and seeds are used as spice. The Ceylon Ironwood with the vernacular name of Nangu is a lofty tree with buttress at the base is known for the drooping pink-coloured young leaves and large fragrant white flowers. This is the National Tree of Sri Lanka and State Flower of Tripura

and Mizoram. This kind of scientific information what I gathered from different sources was of great use during the field inspection of the Higher Officers.

By the time we reached Karumandi Amman River, fresh excreta of Sloth Bear were seen on the rock boulders. It is a common sight to see nearly half-a dozen bears in front of the rest house during the fruiting season of the Phoneix. They used to move around the forest in search of food like honey comb, jungle fruits like amla, jack fruit, wild durian (*Cullenia excelsa*), wild date palm (*Phoenix humilis*), etc. While part of the Karumandi Amman River flows down and joins Manimuthar on the western side, the other part is diverted by the stone wall built across the

river and allowed to join the Nanguneri Canal. Having come all the way to the temple, we prayed the Goddess for a while.

On our return, a few Thitmin (*Podocarpus wallichianus*-Narambili) trees that belong to the family 'Coniferae' grown along the river bank were shown to the CF. The gymnospermous tree is generally tall and evergreen. Generally during the field visit, I used to carry Mr T.R. Somasundaram's Flora book and the Hand Book on 'Trees, Shrubs, Climbers and Herbs of Madras Presidency' by M/S. V.N. Seshagiri Rao and M.H. Krishnaswamy with me for identifying any floral species. It was then decided to return to the rest house.

Orchidarium at Sengaltheri: Orchids like *Anoectochilus elatus* (Ground Orchid), *Anoectochilus regalis* (Marbeld Jewel Orchid), *Bulbophyllum aureum* (Golden Bulb-Leaf Orchid), *Bulbophyllum fishier* (Fisher's Bulb-Leaf Orchid), *Bulbophyllum tremulum* (Tremulous Bulb-Leaf Orchid), *Calypso bulbosa* (Fairy Slipper), *Cypripedium montanum* (Mountain Lady Slipper), *Cymbidium ensifolium* (Golden-thread Orchid or Spring Orchid), *Dendrobium aqueum* (Aqua Dendrobium), *Dendrobium diodon* (The Two Toothed Dendrobium), *Paphiopedilum druryi* (Lady's Slipper Orchid), *Rhynchostylis retusa* (Foxtail Orchid) and *Vanda testaceae* (The Small Flowered Vanda) had been collected from various parts of the Kalakad RF and nearby forests and raised in plastic and clay pots with humus, brick pieces, charcoal and loamy soil as medium and kept hanging from different trees under the shade and the terrestrial orchid had been kept on the ground. Some of the epiphytic orchids had been kept on the tree branches.



Foxtail Orchid



Lady's Slipper Orchid

After trekking for a distance of about a kilo metre, while coming down, almost nearer to the rest house, on the western side, we had formed an 'Orchidarium' (An area dedicated to Orchids) in the midst of the dense forest. After observing some orchids with colourful and beautiful flowers at Kakkachi, I was inspired to form this Orchidarium. With due permission from the WLW covered by an estimate, it was developed with great difficulty.

At the time of the CF's visit to the Orchidarium, I was giving the following details to him. "The family Orchidaceae is said to be the second largest family of flowering plants. It contains about 880 genera and about 26,000 species distributed throughout the world. Orchids are perennial herbs. While many are cultivated as ornamentals for their beautiful and colourful flowers, several have economic values. Some of the interesting Orchids are called Bucket Orchid, Butterfly Orchid, Donkey Orchid, Frog orchid, Jewel Orchid, Ladies' Tresses, Lady's Slipper, Lizard Orchid, Moth Orchid, Man Orchid, Spider Orchid, Sun Orchid and Bird's Nest Orchid". After listening to me, the CF appeared to be impressed and instructed me to have some more collections and maintain them properly. When the CF enquired about the identification of the Orchids, I told him that the identification of the Orchids was done with the help of a Doctor from Palayamkottai who was well versed in Botany, especially in Orchids. With great difficulty, I was able to collect the names of the Orchids from my old pocket note books. As I have already mentioned that I had the opportunity of studying the basic and field Botany only after joining the SFRC.

By the time we reached the rest house, it became 12.30 pm. After lunch, the CF wanted to take rest for a while. At around, 4 pm, with a simple tea, we proceeded to

Kalakad. Within an hour time, we could reach Sivanaperi, the place where the forest complex is situated. I requested the CF to make a visit to the Forest Range Office and he too obliged immediately. While walking towards the office, he was telling that it was known as Nanguneri Range in the earlier days. After occupying the chair meant for the FRO, the CF enquired with the Range Clerk about the offence pendency details. For a while the Clerk started blinking and the very next moment, I opened my pocket note book and listed out the pendency at different stages like pending in the court under trial, for want of 'E' card, sale of seizures, collection of 'C' fees, etc.

Only a few cases were pending for collection of 'C' fees and when the CF questioned me why the cases were pending for collection of 'C' fees, I said that the C & G Notices had been served to the accused and within another 15 days, the fees would be collected and the cases be closed. It seemed that the CF was satisfied with my reply. With some biscuits and tea, the CF and the WLW left for Tirunelveli with warm salute from us.

After their departure, the Range Clerk surprisingly asked me how I could give the details very immediately. I told him that the figures given were from my memory and showed the blank page of the pocket note book. Supposing I had not given the details at that time, the CF would have become very angry and an unpleasant scene would have been created. So, to overcome this kind of unexpected situation, we have to use our presence of mind very tactfully.



V. Sundararaju, IFS., (R)
Former Deputy Conservator
of Forests

MY UNFORGETTABLE EXPERIENCE



Rescue of a tiger

When I was working as Forest Range Officer Top Slip I have witnessed a horrible incident in which a tiger struggled for life whole day. I had attended a religious tribe's goddess festival which was celebrated jointly by Kerala and Tarmilnadu Tribal at the border area near Parambikulam - Aliar water canal (PAP). On 13-05-1986 at about 10 AM we heard alarming trumpeting noise of an elephant followed by a freak crying sound of an elephant female calf which speedily climbed down the steps of 20 feet depth of PAP canal. Within a moment, the tiger quickly pounced on the calf and both fell in to the canal flood which was gushing with high velocity. The tiger was washed away through the small vent in the separation (Tamil Nadu - Kerala Boarder) to Tamil Nadu side and remained trapped in the iron grilled barricade of cave in-tunnel of Parambikulam - Aliyar canal.

The tiger was trapped there and it kept swimming in the water and struggled for life. With the help of local tribes we immediately prepared a wooden-bamboo ladder and dropped it in the canal. The tiger climbed it but the ladder could not withstand its

weight and broke. Till evening we tried our level best to rescue the tiger. In the evening Mr R. Sundararaju, IFS, the then Wild Life Warden came and contacted the Chief Engineer PWD, Chennai and stopped the water flow in the canal. The tiger was able to climb up the steps and happily returned to his home in the forest. Sadly the elephant calf, trapped in Kerala side, died. This episode of rescue of the tiger safety and the effort taken by the staff were appreciated by magazines including Frontline and Sanctuary.

Even though people may think that I had only done my duty as a forest officer, but on that day the amount of agony and struggle undergone by me will ever remain green in my memory.



M. R. Subbaraman,
Former Deputy Conservator
of Forests

“MELTING ICE AND NATURE’S CHANGES - ALASKA”



The word *Alaska* originates from the Aleut word *alaxsxaq*, meaning “the mainland” or “the object toward which the action of the sea is directed.” The Aleut people, also known as the *Unangax* are the Indigenous inhabitants of the Aleutian Islands, located in the western part of the Alaska Peninsula. The name “Aleut” was later given by Russian explorers during their colonization efforts. Today, a small number of Unangax people still live in Russia, as the Aleutian Islands stretch toward Siberia.

On March 30, 1867, Russia sold Alaska to the United States, mainly due to financial difficulties following the Crimean War and the challenges of defending such a remote territory from British expansion. The U.S. purchased Alaska—then known as “Russian America”—for \$7.2 million, which equated to about 2 cents per acre. At the time, many Americans referred to the purchase as “*Seward’s Folly*,” after Secretary of State William H. Seward, who negotiated the deal. *But in reality, Russia lost a very valuable real estate, rich in natural resource, oil and Gas etc.*

Alaska is home to two main Indigenous cultural groups, the Inuit–Unangan (also known as Eskimo Aleut) and the Na-Dene (including the Athabaskan, Eyak, and Tlingit peoples). While English is the dominant language today, many Alaska Native languages are still spoken-though some are endangered. This linguistic heritage is reflected in place names across the state, many of which have Athabaskan origins.

Interestingly, Juneau, the capital city of Alaska, is the only U.S. state capital that is not accessible by road. It can be reached only by air or sea. Despite its political importance, Juneau remains relatively small, with a population of just over 31,000.

My trip to Alaska 5th July 2025:

We proceeded to Anchorage, the largest city in Alaska. With an estimated population of 290,761, it stands as the most populous urban center in the state, followed by Fairbanks, which has around 128,565 residents.

Population of Alaska	7,41,147	Shore Lines	54563 Km
Area	1,51,880 Sq.km	Temperature	-54°C to 34°C
Glaciers	75,109.65 Sq.km	National Parks	9
Total estimated Glaciers	1,00,000	National Forests	2
Coast Lines	10686 Km	State Forests	3

From September to March, Anchorage becomes an ideal spot for witnessing the Northern Lights (*Aurora Borealis*)—a breathtaking natural phenomenon that paints the night sky with ribbons of green, purple, and pink. The city is well connected to the rest of the United States via the Alaska Highway, making it a key transportation hub.

Anchorage is surrounded by more than 50 glaciers, many of which are accessible through day trips or scenic flights. One of the most surreal experiences we had was witnessing the Midnight Sun, where the sun remains visible even at midnight, an unforgettable reminder of Alaska's unique geography and summer daylight.

Rental cars are readily available from Ted Stevens Anchorage International Airport, which makes exploring the surrounding wilderness and scenic routes incredibly convenient.



6th July 2025:

Today was truly special, we took the Alaska Railroad to Seward, a charming port town with a small population but incredible natural beauty. The journey lasted approximately 4 hours and 15 minutes, offering breathtaking views of glaciers, lush green hills, and snow-capped mountains. Along the way, we were lucky to spot a moose, a bald eagle, and mountain goats in their natural habitat.

In Seward, we boarded a glacier cruise that took us through the cold, clear waters surrounded by icebergs and dramatic coastal cliffs. The 4-hour cruise traveled toward the confluence with Alaska Bay (part of the North

Pacific Ocean), and it was nothing short of magical. We spotted a Humpback Whale breaching in the distance, sea lions lounging on rocky outcrops, playful sea Otters floating on their backs, and once again, the majestic Bald Eagle soaring overhead. The highlight of the cruise was seeing a pod of five killer whales (Orcas) gracefully glide through the water, a rare and unforgettable moment.

Our return train departed Seward at 6:00 PM and brought us back to Anchorage by 10:00 PM, capping off an incredible day filled with nature, wildlife, and the wonders of Alaska's wilderness.

7th July 2025:

We proceeded to Talkeetna; a quaint town located about 113 miles north of Anchorage. Though small with a population of just 1,090, Talkeetna serves as a gateway to some of Alaska's most incredible glacier flights. From here, we boarded a scenic flight that took us soaring over vast glacial fields, with an unforgettable landing on the glacier itself. Standing on the ice, surrounded by towering snow-covered peaks, was nothing short of exhilarating. We spent about 15 minutes on the glacier, with spectacular views of Mount McKinley, also known as Denali, the tallest peak in North America at 20,310 feet.

From the air, we could see signs of glacial melting, including visible landslides and crevasses, a stark reminder of the ongoing effects of climate change. The town itself is surrounded by expansive wetlands, creating a contrast between lush lowlands and icy highlands.



As we continued our journey from Anchorage to Denali, we noticed a striking and somewhat haunting sight: miles and miles of dried, dead Spruce trees lining both sides of the road for nearly 100 miles, covering what seemed to be thousands of hectares. At first glance, we assumed it was the aftermath of a forest fire, but upon reaching Denali, I decided to look into the real cause.

It turns out that these trees have succumbed to an ongoing outbreak of “Spruce beetle” infestation, a major ecological issue in Alaska. Since 2016, approximately 9,105 square kilometers (or 2.25 million acres) of forests have been affected. The infestation is now in its seventh year, and another disease called “spruce bud blight” is also contributing to the decline of these trees.



8th July 2025:

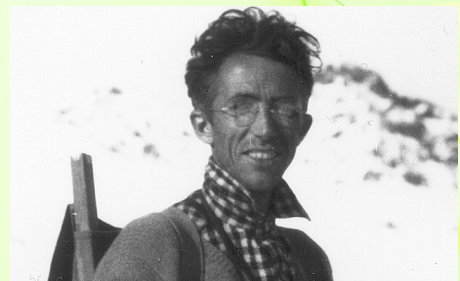
We visited the breathtaking Denali National Park, located about 234 miles from Anchorage. Travel by private vehicle is allowed up to 15 miles into the park, ending at the Savage River, which is a popular starting point for hikers. Beyond this point, access is restricted to park buses to preserve the delicate wilderness ecosystem.

Covering an expansive 24,464 square kilometers, Denali National Park is not only a natural wonder but also a site of deep historical and ecological significance. It was established by the U.S. Congress in 1917, initially under

the name Mount McKinley National Park, with the primary goal of protecting the Dall sheep from overhunting.

The park's creation was largely driven by Charles Sheldon, a conservationist and hunter from Vermont, and Harry Karstens, a legendary outdoorsman and dog musher. During the long winters, they developed the idea of creating the world's first national park focused on wildlife conservation, and after nearly a decade of advocacy, their vision was realized.

Despite its protected status, the Dall sheep population experienced a noticeable decline in the 1930s, and wolves were initially blamed. This led to one of the most significant chapters in wildlife management history “the Wolf-Sheep Controversy”.



Adolph Murie, who is often called “Denali’s Wilderness Conscience.” Born in 1899, Adolph first came to Alaska in 1922 to assist his half-brother Olaus Murie in a caribou study in the Brooks Range. The brothers even attempted to crossbreed wild caribou with domestic reindeer to enhance livestock traits.

After earning his Ph.D. in ecology from the University of Michigan in 1929, Adolph began a distinguished career with the National Park Service (NPS). He conducted wildlife studies across multiple parks, including groundbreaking research on coyote ecology in Yellowstone.

In 1939, the NPS assigned Murie to study the relationship between wolves and Dall sheep in Denali (then Mount McKinley) to resolve the ongoing debate. Over two years, Murie meticulously studied not only wolves and sheep but also their interactions within the broader ecosystem. His research revealed that the decline in sheep populations was

actually due to harsh late-winter conditions, not wolf predation. He argued that predators played a vital role in maintaining the balance of the ecosystem.

Murie published his findings in 1944 as a government bulletin titled *The Wolves of Mt. McKinley*, which became a landmark ecological study. It was one of the first in-depth investigations into wolf behavior and their role in nature, laying the foundation for modern ecological management in national parks.

Today, Denali National Park stands not only as a symbol of natural beauty but also as a testament to the evolving philosophy of conservation, from predator control to the protection of entire ecosystems.



9th July 2025:

We visited Portage Glacier, located about 70 miles southeast of Anchorage, within the beautiful Chugach National Forest. This glacier is one of the most accessible in Southcentral Alaska and seeing it up close was a truly memorable experience.

To reach the glacier, we took a 90-minute cruise across Portage Lake, allowing us to enjoy a close-up view of the glacier's towering vertical face. Along the way, we saw icebergs floating serenely in the water, chunks that had calved off the glacier itself. The cruise, which includes a narrated tour, departs near the Begich, Boggs Visitor Center and is an ideal way to explore the icy landscape. The glacier cruise season typically runs from late May to early September.

One of the interactive highlights was at the visitor center, where dense glacier

ice collected from the lake, is sometimes displayed so visitors can touch and feel it. It's a rare opportunity to physically connect with thousands of years of natural history.

For those who prefer to explore by land, there are several nature trails nearby offering breathtaking views of the glacier, the iceberg-dotted lake, and the lush surrounding valley. The nearby town of Whittier provides another route to the glacier, accessible via the Anton Anderson Memorial Tunnel, a 2-mile-long one-way tunnel shared by cars and trains. Passing through it was an adventure in itself and added to the thrill of the day.

Portage Glacier is not just scenic, it's one of Alaska's most visited attractions, offering both accessibility and awe-inspiring beauty in a compact, unforgettable experience.

10th July 2025 (Glacier Melting):



Today, we explored the Matanuska Glacier, a stunning valley glacier that feeds the headwaters of the Matanuska River. Located about 103 miles from Anchorage and around 60 miles south of Palmer, this glacier is one of the few in Alaska that is easily accessible by road, making it a perfect day trip destination.

Our visit took us to Glacier View, a small yet active community along the Glenn Highway National Scenic Byway, stretching between Mileposts 100 to 120. Although not an officially recognized town, Glacier View is the heart of glacier-related tourism in the area, bustling with local businesses offering a wide range of adventure tours and outdoor activities.

The Matanuska Glacier is the largest road-accessible glacier in Alaska, and it truly dominates the landscape. Visitors can either trek across the glacier independently or, for a safer and richer experience, hire a professional guide. If We opted for a guided trek, which allowed us to learn more about the glacier's formation, features, and the surrounding Chugach Mountains.

In addition to trekking, the region offers white-water rafting on the Matanuska River, zip-lining, ATV tours, and even luxury camping—or “glamping,” as it's fondly known in Alaska. The views were spectacular: panoramic scenes of ice fields, rugged peaks, and winding rivers.



Interestingly, although the glacier is still active, constantly moving downhill under the force of gravity its terminus (endpoint) has shown relatively little movement over the past 30 years. The ice takes roughly 250 years to form in the upper reaches of the glacier and eventually travels to the terminus, making it a powerful timeline of nature's gradual work.

However, the melting of glaciers carries significant environmental impacts. During the melt season, soil erosion becomes a serious concern. The melting snow and ice wash away the soil, exposing bare rock underneath. This leads to both landscape transformation and ecological disruption.

A surprising statistics I came across: the annual transport of gravel, a byproduct of glacial melt and erosion, from this region to other parts of the U.S. is estimated to be as high as 5.85 million tons. The material that isn't used domestically eventually makes its way to the sea, contributing to sediment build up in downstream ecosystems.

The trip to Matanuska Glacier was not only adventurous but also deeply educational, leaving us with a stronger appreciation for both the majesty and fragility of Alaska's glaciers.

Reference:

1. RV Alaska campgrounds.
2. Anchorage, Fairbanks, Juneau, Valdez Guides



V. Jayaprakasam, MSc, PM (RS & GIS),
Former Assistant Conservator
of Forests

ALASKA - GREY WOLF



Alaska is home to an estimated 7,000 to 11,000 healthy gray wolves (*Canis lupus*) covering roughly 85 percent of the state. They thrive in diverse habitats from coastal rainforests to arctic tundras, playing a vital ecological role as apex predators.

A TRIBUTE TO MY FRIEND - M. K. KARUNANITHY



M.K. Karunanidhy, DFO (Retd)

31 October 1930 - 13 July 2026

Aged 95 years and 8 months

It is with profound sorrow that I pay my heartfelt tribute to Thiru M. K. Karunanidhy, a distinguished Forest Officer and an affectionate elder, who passed away on 13 July 2026.

He was born at Minnur, Ambur Taluk, in present-day Thirupattur District. His father, Shri Kuppusamy Naidu, was a close friend of my father, and our families have shared a warm relationship spanning three generations.

After completing his School Final examination at the age of 17, he joined the Andaman Forest Department as a Forester in 1947, when the Boy Service Scheme was in vogue. After undergoing forestry training in West Bengal, he rose through the ranks to become a Forest Range Officer and later Divisional Forest Officer (Social Forestry). He served with distinction in the Andaman Forest Department and later in the Tamil Nadu Forest Department at Thiruvallur, Rasipuram, Thirupattur, Tambaram, Krishnagiri and Thiruvannamalai.

An outstanding feature of Thiru Karunanidhy's career was his remarkable 41 years of dedicated service in the Forest Department. Such an exceptionally long tenure is a rare distinction in the history of the department and stands as a testimony to his unwavering commitment, hard work and devotion to forestry. Few officers have had the privilege of rendering such long and dedicated service.

He was widely respected for his integrity, sincerity and exceptional knowledge of timber species. He successfully managed the Andaman Forest Mill and the Andaman Timber Depot at Chennai. His dedication earned the confidence and appreciation of several senior officers, including Thiru T. Achaya, IFS.

Thiru M. K. Karunanidhy was a compassionate administrator who believed in correcting rather than punishing his staff. He was deeply committed to their welfare and served as President of the Forest Department Association, always working for the well-being of employees. After retirement, he continued his service to the forestry fraternity as Joint Secretary of the Tamil Nadu Association of Senior Professionals of Forestry and Environment (TASPEF) for six years.

To me, he was more than a senior officer—he was an affectionate elder brother who always addressed me as “Thambi.”

One of his cherished wishes was to worship Lord Kashi Viswanath at Varanasi. Although his health was fragile, I was fortunate to fulfil that wish in 2019 by accompanying him on the pilgrimage. We had the divine darshan of Lord Viswanath, witnessed the sacred Ganga Aarti and enjoyed a memorable boat ride on the holy Ganga. My friend Shri Madan Ram kindly arranged VIP darshan for us. I also had the privilege of taking him to Rameswaram, Dhanushkodi and the ancient Uthirakosamangai Temple. It gives me immense satisfaction that I could help fulfil his long-cherished spiritual aspirations.

Our friendship extended across three generations. Even today, my son, who lives in the United States, shares a close relationship with his daughter, reflecting the enduring bond between our families.

His passing is a great loss to the forestry fraternity and to all who knew him. I have lost a beloved elder brother and lifelong well-wisher. His simplicity, humility, compassion, devotion to duty and concern for the welfare of others will always remain an inspiration.

I pray that the Almighty grants eternal peace to his noble soul and gives strength and solace to his bereaved family.

May his noble soul rest in eternal peace. Om Shanti.



V. Sambasivam, IFS (Retd.)

Houston, USA

விலங்குகளின் கொம்பு



“கொம்புளதற்கு ஐந்து குதிரைக்கு பத்துமுழம்
வெம்புகரிக்கு ஆயிரம்தான் வேண்டுமே - வம்புசெறி
தீங்கினர்தம் கண்ணில் தெரியாத தூரத்து
நீங்குவதே நல்ல நெறி.” (நீதிவெண்பா 20)

இதன் பொருள், கொம்புள்ள விலங்குகளிடமிருந்து ஐந்து முழம், குதிரையிடமிருந்து பத்து முழம், யானை போன்ற விலங்குகளிடமிருந்து ஆயிரம் முழம் விலகி நிற்க வேண்டும்; அதுபோல தீங்கு செய்பவர்களிடமிருந்து கண் காணாத தூரத்தில் விலகி இருப்பதே நல்ல நெறி என்பதாகும்.

பழமையான தமிழ் இலக்கியப் பாடல்களிலும் கொம்புள்ள விலங்குகளிடமிருந்து எவ்வளவு தூரம் பாதுகாப்பாக இருக்க வேண்டும் என்ற உவமை வருகிறது. அதில், கொம்புள்ளவற்றைச் நெருங்கும் போது எச்சரிக்கையாக இருக்க வேண்டியதைக் குறிப்பிடுகிறது.

கொம்புகளின் பயன்

விலங்குகளின் கொம்புகள் என்பது தற்காப்பு, போட்டி, மற்றும் சில இனங்களில் கவர்ச்சி ஆகியவற்றுக்காக இயற்கை வழங்கிய முக்கிய உடல் அமைப்பு. கொம்புகளின் வடிவமும் அமைப்பும் விலங்கினத்துக்கு விலங்கினம் மாறுபடும்; மாடு, எருமை, ஆடு, மான், காண்டாமிருகம் போன்றவற்றில் அவை வெவ்வேறு பயன்பாடுகளைக் கொண்டுள்ளன.

வட்டார வழக்குகளில் கொம்பு

தென் மாவட்டங்களில் ‘கொம்பு’ என்ற சொல் விலங்குகளுக்கு மட்டும் உரித்தானவை. ஆனால் வட மாவட்டங்களில் மரத்தின் கிளைகளில் உண்டாக்கப்பட்ட குச்சிகளுக்கும்/சிறிய இலகுவான தடிகளுக்கும் பொருந்தும் தென் மாவட்டங்களில் மரத்தினால் உண்டாக்கப்பட்ட தடிகளுக்கு /குச்சிகளுக்கு கம்பு என்றே அழைக்கப்படுகிறது .

உலக வழக்கில் கொம்பு

“Horn of Africa” என்பதற்கு தமிழில் “ஆப்பிரிக்காவின் கொம்புப் பகுதி” அல்லது “ஆப்பிரிக்காவின் கொம்பு” என்று கூறப்படுகிறது.

ஆப்பிரிக்கக் கொம்பு என்றால் என்ன?

ஆப்பிரிக்க கண்டத்தின் வடகிழக்குப் பகுதியில், கடலுக்குள் கொம்பைப் போல நீண்டு காணப்படும் தீபகற்பப் பகுதியே ஆப்பிரிக்கக் கொம்பு என அழைக்கப்படுகிறது. இதில் அடங்கும் நாடுகள்

1. சோமாலியா (Somalia)
2. எத்தியோபியா (Ethiopia)
3. எரித்திரியா (Eritrea)
4. ஜிபூட்டி (Djibouti)

ஆப்பிரிக்க வரைபடத்தைப் பார்க்கும்போது, இந்தப் பகுதி காண்டாமிருகம் அல்லது பிற விலங்குகளின் கொம்பைப் போல கடலுக்குள் நீண்டு இருப்பதால் “ஆப்பிரிக்க கொம்பு” என்று பெயர் பெற்றது.

கொம்பேறி மூக்கன்

பாம்பில் ஒருவகை பாம்பிற்கு “கொம்பேறி மூக்கன்” என்ற பெயர்... அது புதர் அல்லது செடியின் உச்சிக்கு வேகமாக ஏறும் அதனால் கொம்பேறி மூக்கன் என்ற பண்பு பெயரால் சுட்டி அழைக்கப்படுகிறது.

திருக்குறளில்

“நுனிக்கொம்பர் ஏறினார் அஃது இறந்து ஊக்கின் உயிருக்கு இறுதியாகிவிடும்”

(வலி அறிதல்சு என்ற அதிகாரத்தில் குறள் எண் 476)

இதன் பொருள் மரத்தின் உச்சியில் ஏறி நின்றவர் மேலும் ஏறக்கூடாது அதை மீறி ஏறமுயன்றால் கீழே விழுந்து உயிர் துறப்பர் சுருங்கச் சொன்னால் தன்னுடைய திறன் சூழ்நிலை அறிந்து செயல்பட வேண்டும் என்று பொருள் . இந்தக் குரலிலும் கொம்பு என்ற சொல் பயன்படுத்தப்பட்டுள்ளது

சீவக சிந்தாமணியில் மான் கொம்பு

உழையினும் உச்சிக் கோடு கலங்குதற்கு போதே விழைவு அற விதிர்த்து வீசி, விட்டெறிந்திடுவது ஒப்பக் கழலவர் உள்ளம் அஞ்சிக் கலங்குமேல், அதனை வல்லே மழையினின் நீக்கி இட்டு வண்கண்ணார் ஆவர் அன்றே.

(பாடல் எண்:2301.-மணமகள் இலம்பகம்)

பொருள்

மான்கள் தனது முதுமைக் காலத்தில் அசையும் கொம்புகளைப் பற்றி மனம் கலங்கி வருந்தும். ஆயினும் அப்போதே தன் மனதைத் திடப்படுத்திக் கொண்டு அக்கொம்புகளை தானாக அசைத்துப் பிய்த்து வீசியெறிந்து விடும். அதைப் போன்று வீரர்கழல் அணிந்த வீரர்கள் போரில் தமது உடல் மேலும், உயிர் மேலும் ஆசை கொண்ட ஆசைகளை மின்னல் வேகத்திலே தூக்கி எறிந்து அச்சத்தை இருந்த இடம் தெரியாமல் விரட்டிவிடுவார்கள். அப்படி அகற்றுபவர்களே சிறந்த வீரர்கள் என அழைக்கப்படுவர். காப்பியங்களில் சீவக சிந்தாமணி மாத்திரம் தான் மான் கொம்பு பற்றிக் குறிப்பிடுகிறது .

ஜோதிடத்தில் கொம்புள்ள விலங்குகள்

ஜோதிடத்தில் பன்னிரெண்டு ராசிகளில் மூன்று ராசிகளில் கொம்புள்ள மிருகங்கள் அடையாளப் படுத்தப்படுகிறது

1. மேஷ ராசிக்கு(Aries) கொம்பு உடைய ஆடும்,
2. ரிஷப ராசிக்கு (Taurus)கொம்புள்ள காளையும்,
3. மகர ராசிக்கு (Capricorn) கடல் மிருகம் அதாவது கொம்புடைய ஆட்டின் தலையும் உடம்பு மீன் வடிவமும் அடையாளமாக உள்ளன

தேசிய விலங்காக கொம்புள்ள விலங்குகள்

பாகிஸ்தான் நாட்டின் தேசிய விலங்கு மார்க்கர் எனும் காட்டு ஆடு (Markhor) . அமெரிக்க நாட்டின் தேசிய விலங்கு அமெரிக்க காட்டெருமை (Bison) .

பசு, நேபாளத்தின் தேசிய விலங்கு. பூட்டான் நாட்டின் தேசிய விலங்கு “தகின்” (Takin) கொம்புள்ள ஆடு மற்றும் மாடு கலந்தது போல் இருக்கும்.

மனிதரும் விலங்கின் கொம்புகளும்

இயற்கை வளங்களையும், காடுகளையும்,

விலங்கினங்களையும்

பாதுகாக்க

வனப்பாதுகாப்பு சட்டங்கள் வருவதற்கு முன்பு மனிதர்கள் குறிப்பாக ஜமீன்தார்கள், நிலப்பிரப்புகள், உயர் அதிகாரிகள் தடையின்றி காட்டில் வேட்டையாடுவதை வீரத்தின் அடையாளமாகவும், பொழுது போக்காகவும் பெருமையாக கொண்டிருந்தனர். வேட்டையாடிய மிருகங்களை பதப்படுத்தி அவைகளின் தலைகளை வீட்டில் வைப்பது பெருமையாகவும் கருதினர். அது அவர்கள் வாழும் இடத்திற்கு அழகாக இருந்தது அதன் கூடவே அதை சுட்ட துப்பாக்கியும் அருகில் இருக்கும் அதில் புலியின் தலையும் மானின் கொம்போடு கூடிய தலையும் பங்கு பெற்றிருக்கும் புலியின் தலை அச்சுறுத்தும் மானின் தலை தன் அழகை வெளிப்படுத்தும். இப்போது இவைகளை காண்பது அரிது.

“கொம்பன்” என்ற சொல்

மனிதரில் மூர்க்க குணம் உடையவரையும் அடங்காதவர்களையும் அவனை கொம்பன் என்று அழைக்கும் வழக்கம் உண்டு .. அவர்களும் தன்னை கொம்பன் என்று அழைப்பதில் பெருமை கொள்வர். கேரளத்தில் “கொம்பன்” என்று சொன்னால் ஆண் யானையை குறிக்கும். ஆகவே கொம்பன் என்ற சொல் விலங்குகளுக்கும் மனிதர்களுக்கும் நடை முறையிலும் இலக்கியத்திலும் பயன்பாட்டில் உள்ளது.

“LOCKING HORNS” (தலையிட்டு மோதுதல்)

என்பது ஆங்கிலத்தில் இரண்டு பொருள்களில் பயன்படுத்தப்படுகிறது.

1. நேரடி பொருள்

இரண்டு கொம்புள்ள விலங்குகள் (மான்கள், ஆடுகள், காளையக் கோழிகள் போன்றவை) தங்கள் கொம்புகளை ஒன்றோடொன்று மோதிக்கொண்டு சண்டையிடும் போது கொம்புகள் பிண்ணிக்கொள்ளும் அதை “locking horns” என்பார்கள்.

2. உருவகப் பொருள் (Figurative meaning)

இரண்டு நபர்கள் அல்லது குழுக்கள் கடுமையான வாக்குவாதம், கருத்து வேறுபாடு காரணமாக மோதலில் ஈடுபடுவதை குறிக்கும்.

எடுத்துக்காட்டு

1) The two stags were locking horns in the forest. → இரண்டு ஆண் மான்கள் காட்டில் கொம்புகளைப் பூட்டிக்கொண்டு

சண்டையிட்டன. (நேரடி பொருள்)

2) The two politicians locked horns over the new policy. → புதிய கொள்கையைப் பற்றி இரண்டு அரசியல்வாதிகள் கடுமையாக மோதிக்கொண்டனர். (உருவாகமாக)

கொம்பு உள்ள சில விலங்கினங்கள்

மான் வகைகள், எருமை, பசு, ஆடு, காண்டாமிருகம்.

யானைத் தந்தம் கொம்பு போல் உள்ளது அவை கொம்பு அல்ல பற்களின் நீண்ட வளர்ச்சியாகும் அதனால் அவற்றை கொம்பு என்று கருதலாகாது.

மான் கொம்பு, மாடுகளின் கொம்பு காண்டாமிருகக் கொம்பு அமைப்பு

மான் கொம்பு, மாடுகளின் கொம்பு, காண்டாமிருகத்தின் கொம்பு — மூன்றும் ஒரே மாதிரி தோன்றினாலும் அமைப்பு வேறு. மான் கொம்பு எலும்பு அடிப்படையிலான கிளைபோன்ற அமைப்பு; மாட்டுக் கொம்பு மையத்தில் எலும்பு முளை மீது கெரட்டின் உறையால் ஆனது. காண்டாமிருகக் கொம்பு முழுவதும் கெரட்டின் நாள்களால் ஆனது. உள்ளே எலும்புக் கரு இல்லை

அடிப்பகுதி (base):

கொம்பின் தொடக்கம் தலை தோல் அணுகுமிடத்தில் பிணைந்திருக்கும்; இதுவே கொம்பை இயக்கும் சிற்றங்குகள் உள்ள பகுதி. [1]

நடுப்பகுதி (shaft):

கொம்பின் மையமான நீளமான பகுதி, திடமாகவும் வளைந்த வடிவமாகவும் இருக்கும்.

உச்சிப்பகுதி (tip):

கொம்பின் முனை கூர்மையாகக் கிடக்கலாம் அல்லது துப்பிச் சரிவாக முடிந்திருக்கும்; போர்ப்பிடிக்கும் போது இவ்வான் முக்கியம்.

கொம்புகளின் பயன்பாடு

விலங்குகள் கொம்புகளை எதிரிகளைத் தடுக்கவும், தங்களைப் பாதுகாக்கவும் பயன்படுத்துகின்றன. சில இனங்களில் ஆண் விலங்குகள் இனப்பெருக்க காலத்தில் போட்டியாளர்களுடன் மோதவும், தங்கள் வலிமையை வெளிப்படுத்தவும் கொம்புகளைப் பயன்படுத்துகின்றன. கொம்புகள் என்பது விலங்குகளின் பாதுகாப்புக்கும் உயிர்வாழ்வுக்கும் உதவும் இயற்கை அமைப்பு; அதேசமயம் சில இனங்களில் அவை அழகும் அடையாளமும் ஆகும்.

கொம்புகள் பெரும்பாலும் தற்காப்பு, எதிரியை அச்சுறுத்தல், இனப்போட்டி, துணை ஈர்ப்பு, ஆகியவற்றுக்குப் பயன்படுகின்றன. மான்களின் கொம்புகள் ஆண்டுதோறும் விழுந்து புதிதாக முளைப்பது ஒரு தனிச்சிறப்பு. காண்டாமிருகத்தின் கொம்பு அந்த விலங்கின் அடையாளமாகவே கருதப்படுகிறது. ஆண் காண்டாமிருகங்கள் பொதுவாக கொம்புக்களை பிற ஆண்களுடன் காட்சி க்காகவும் (display) மற்றும் போராட்டத்திற்கும் பயன்படுத்துகின்றன.

உணவுச் செயல்கள்:

கொம்புகள் மண்ணை தோண்டுதல், தாவரக் கிழங்குகளை மிதவை செய்வது போன்ற பயன்களில் உதவுகிறது .

- தன்முறை செயல்கள்: சில வகையில் கொம்பின் வடிவம் அதன் உயிரியல் சூழ்நிலைக்கு (உதாரணம்: மின்சார மூட்டம், உணவு அமைப்பு) ஏற்ப மாறிவரும். கொம்பு உள்ள விலங்குகள் பொதுவாக தாவர பட்சியாக (Herbivorous) இருக்கின்றன

விலங்குகளின் கொம்புகளை பொதுவாக இவ்வாறும் வகைப்படுத்தலாம்:

- நிரந்தரக் கொம்புகள்

மாடு, ஆடு, எருமை போன்றவற்றில் காணப்படும். இவை பெரும்பாலும் வாழ்நாள் முழுவதும் இருக்கும்.

கலைக்கொம்புகள்

மான்களில் காணப்படும். இவை ஒவ்வொரு ஆண்டும் உதிர்ந்து மீண்டும் வளரும்.

எண்ணிக்கையின் அடிப்படையில் ஒற்றைக் கொம்பு:

இந்தியா காண்டாமிருகத்தில் ஒரே ஒரு கொம்பு மட்டும் இருக்கும்.

இரட்டைக் கொம்பு:

சில விலங்குகளில் இரண்டு கொம்புகள் இருக்கும்; உதாரணமாக சில மான் ஆடு , மாடு வகைகள், ஆப்பிரிக்க காண்டாமிருகம்

பல கொம்புகள்: (Four Horned Antelope)

சில சிறப்பு மானினங்களில், நான்கு கொம்புகள் உள்ள வகையும் உண்டு.

காண்டாமிருக கொம்புகளின் வகைகளின் சிறப்பு

இந்திய (ஒரே கொம்பு) மற்றும் ஆப்பிரிக்க (இரு கொம்புகள்) வகைகள் வேறுபடுகின்றன.

- முன்கொம்பு (anterior) மற்றும் பின்கொம்பு (posterior):

இரண்டு கொம்புகளை கொண்ட ஆப்பிரிக்க வகைகளில் முன் கொம்பு சிறிதாகவும் பின் கொம்பு பெரியதாகவும் இருக்கும்.

- ஒற்றைக்கொம்பு (one-horned): இந்திய கசிரங்கா போன்றவைகள்; மேல்முனையில் ஒரு கொம்பு மட்டும் இருக்கும்.

- இரட்டை கொம்பு (two-horned): ஆப்பிரிக்க கருப்பு மற்றும் வெள்ளை காண்டாமிருகங்கள்; இரண்டு தனித்தன்மை கொண்ட கொம்புகள் இருக்கும்.

வளைந்த கொம்புகள்:

சில வகைகளில் கொம்பு வளைந்தோ அல்லது கீழ்ப்புறம் நெறுங்கோல் கொண்டோ இருக்கும்; இது போர்க்கொள்ளும் நடைமுறைக்கு உகந்ததாக இருக்கும்.

பாலின வேறுபாடு:

ஆண் மான்களுக்கு மட்டுமே கொம்பு இருக்கும் பெண் மான்களுக்கு கொம்பு இல்லை அல்லது சிறியது. ஆனாலும் மாடுகளில் இரு பாலிலும் கொம்பு இருக்கலாம்.

கொம்புகளின் மருத்துவ பயன்கள்

மரபு மருத்துவத்தில் சில மிருகங்களின் கொம்புகள் (எ.கா. மான், காண்டா வகைகள்) மற்றும் கால்நடை கொம்புகள் பல மருந்தியல் பயன்பாடுகளுக்கு பயன்படுத்தப்பட்டதாகப் பதிவுகள் உள்ளன, ஆனால் இதற்கு வரையறுக்கப்பட்ட அறிவியல் ஆதாரம் குறைவு. காண்டாமிருகக் கொம்புகள் போன்றவற்றின் உறுப்புகள் மருத்துவ பயன்கள் அறிவியல் சோதனைகள் இல்லாதவை என்று ஆய்வுகள் குறிப்பிடுகின்றன. பல பயன்பாடுகளும் பாரம்பரிய சான்றுகளுக்கு மையமாக உள்ளன, வனவிலங்கு பாதுகாப்பு சட்டம் மற்றும் ஒழுங்கு: முறை சில விலங்கு பொருட்களுக்கு (மான் கொம்பு, சில அரிய விலங்குகளின் உறுப்புகள்) அரசு தடைகளும் கடுமையான விதிகளும் அமுல்படுத்தி உள்ளன. அதனால் அதன் வைத்தியம், வணிகம் அல்லது ஊரக சந்தை / உலக சந்தையில் வணிகம் சட்டபூர்வ விசாரணைக்கு உட்பட்டதே ஆகும்.

எச்சரிக்கை

பாதுகாப்பு: மரபு முறையில் பயன்படுத்தப்படும் சில விலங்கு உறுப்புகள் நச்சுத்தன்மை, கழிவுகளும் அல்லது தொற்று ஆபத்துகளையும் கொண்டதாக இருக்கலாம்

மருத்துவ முறையாகப் பயன்படுத்துவதற்கு முன் மருத்துவ நிபுணரை ஆலோசனை பெற வேண்டும்

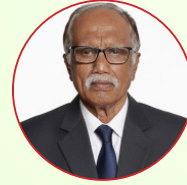
சட்டம் மற்றும் தரம்: அரிய விலங்குகளின் உறுப்புகள் தொடர்பான வணிகம் பல நாடுகளில் தடைசெய்யப்பட்டுள்ளது; விலங்குகளை கொண்டு உறுப்புகள் எடுத்தல் அல்லது வாங்குதல் சட்டவிரோதமாக கருதப்படுகிறது

பயனுள்ள மாற்று வழிகள்:

உண்மையான சிகிச்சைக்கு அறிவியல் ஆதாரம் உள்ள மூலிகை மருந்துகள் மற்றும் வைத்திய ஆலோசனைகள் விருப்பமாக பரிந்துரைக்கப்படுகிறது .

விலங்குகளை கையாள்வதில் பல கட்டுப்பாடுகள் இருந்தாலும் இன்னும் அதை வலிமைப்படுத்தவேண்டும் .

விலங்குகள் நம்மைப் போன்ற உயிரினங்கள் அவற்றின் அடிப்படை உரிமைகளை பாதுகாக்க வேண்டியது நமது கடமை.



சா. சுடலைமுத்து .BSc.(Ag)M.A.LLb
மேனாள் மேலாளர் -பாரத ஸ்டேட் வங்கி

KNOW YOUR BACKYARD HERB

Boerhavia diffusa Linn.:

Common Name: Tamil: Mukarattai
Keerai; English: Spreading Hogweed



Botanical medicinal systems in various forms and formulations pervaded in this world well before the present Allopathic medical system into be known universally as Modern Medicine. “*Food is Medicine; Medicine is Food*” is an age-old Tamil quote. “*Let food be thy medicine,*” said Hippocrates. In Tamil Nadu, the ancient Tamil traditional Siddha medicinal system is also based on this foundation. The Mukuratai Keerai, known outside Tamil Nadu as Punarnava, literally meaning “that which renews or rejuvenates.” Punarnava is a much-celebrated herbaceous plant, for its powerful medicinal properties and its incredible ability to regenerate. Its name comes from the Sanskrit words ‘Punar’, meaning “again” and ‘Nava’ meaning “new,” perfectly capturing the essence of

the plant. This herb is one such example to prove that food is medicine and medicine is food. The leaves of *Boerhavia diffusa* are often used as a green vegetable in many parts of India.

Boerhavia diffusa Linn., a flowering herb, belonging to family Nyctaginaceae, with pantropical distribution is widely used in traditional Ayurvedic, Unani, and folk medicine systems. Ancient texts like **Charaka Samhita** and **Sushruta Samhita**, dating back to 1000 BCE, have documented Punarnava’s usage, highlighting its diuretic and anti-inflammatory properties, especially for treatment of kidney stones (Mutrashmari) and dropsy (Jalodara). This creeping, perennial, much-branched herb with fusiform roots is also credited with its excellent antiaging and antioxidant properties.



Plants under family Nyctaginaceae are popularly known as the **four o’clock family** with about 33 genera and 290 species of flowering plants widely distributed in tropical and subtropical regions. Some species are also reported in the temperate regions. The popularity of this herb and its wide use internationally, can be known through its common names listed

herein. In Chinese it is **Huang Xi Xin**. In French (French Guiana) it is called as **Ipecacuanha de Cayenne**. In Portuguese language in Brazil, it is **Agarra-pinto or Celidônia or Pega-pinto**. In Spanish it is known as **Hierba de Cabro**. Its Urdu name is **Tukhm-i-Ispast**, an indication that this herb is used in Unani system of medicine too.



This commonly ignored backyard weed like plant has been researched by Krutika P. Mokal & Rashmi A Bagul from the Dr Naikwadi College of Pharmacy, Jamgoan, Sinnar, Nashik, Maharashtra and published a very comprehensive review of *B. diffusa* herb's Phytochemistry and Pharmacological activities in the Journal of Advance And Future Research, (JAAFR / Volume 4, Issue 1 January 2026).

As per their research findings, *B. diffusa* is "well-known for its rejuvenating, anti-inflammatory, diuretic, hepatoprotective, and antioxidant properties. Bioactive compounds like punarnavine, alkaloids, flavonoids, lignans, steroids, and glycosides, discovered through phytochemical research, contribute to *B. diffusa*'s diverse therapeutic potential. Various experimental and clinical studies have demonstrated its efficacy in treating liver disorders, kidney dysfunction, inflammation, edema, respiratory diseases, and microbial infections. Due

to its wide pharmacological spectrum and minimal toxicity, *Boerhavia diffusa* remains an important subject of modern pharmacological research"

Boerhavia diffusa can be propagated through its seeds, known as Punarnava seeds or Tarvine seeds. These seeds are small and should be sown in well-draining soil. To promote seed germination the substrate soil needs be kept moist but not water-logged. This versatile herb can be grown either directly in ground or in containers.

Commercial cultivation is undertaken in the states of Rajasthan, Madhya Pradesh and Gujarat, with per acre cost of cultivation ranging from Rs 40,000 to Rs. 70,000/-. Net Profit per acre ranging from Rs 40,000/- to 1,30,000/- within the crop maturity duration of 4 to 6 months. Prices vary with quality, moisture content, fresh herb or dried herb, or roots. Profit depends primarily on securing a buyer before planting and producing clean, properly dried material. Major buyers include, Dabur India, Himalaya Wellness Company, Patanjali Ayurved, Natural Remedies, Aimil Pharmaceuticals and regional herbal raw material traders in Jaipur, Indore, Ahmedabad, and Chennai.

It is time that we use this plant to its fullest potential and as a green vegetable in our diet too.



V. Prabhakaran, IFS., (R)

Former Additional Principal
Chief Conservator of Forests

Photos Courtesy Dr. K. Ravi Kumar
Biodiversity Expert, TBGPCCR, Chennai

ஜான் மியர்: தேசியப் பூங்காக்களின் தந்தை!



ஜான் மியர்

இயற்கையின் அருமையையும், அதன் ஆன்மீகப் பெருமையையும் மனித குலத்திற்கு உணர்த்திய சில பேரறிஞர்களில் மிக முக்கியமானவர் ஜான் மியர் (John Muir). ஸ்காட்லாந்தில் பிறந்து அமெரிக்காவில் வாழ்ந்த அவர், இயற்கையைப் பாதுகாக்கும் இயக்கத்திற்கு ஒரு புதிய திசையைக் காட்டியவர். “தேசியப் பூங்காக்களின் தந்தை” என்று போற்றப்படும் மியர், காடுகள், மலைகள், ஆறுகள் போன்ற இயற்கை வளங்கள் வெறும் வளங்களல்ல; அவை மனித ஆன்மாவை வளர்க்கும் புனிதத் தலங்கள் என்று உலகிற்கு எடுத்துக்காட்டினார்.

1838-ஆம் ஆண்டு, ஸ்காட்லாந்தின் டன்பார் பகுதியில் பிறந்த மியர், சிறுவயதிலேயே இயற்கையுடன் நெருக்கமாக இருந்தார். கடற்கரை, பறவைகள், புல்வெளிகள் போன்றவை அவரின் முதல் ஆசிரியர்களாக இருந்தன. 11-வயதில் அவர் குடும்பத்துடன் அமெரிக்காவின் விஸ்கான்சினுக்குக் குடிபெயர்ந்தார்.

அங்கு சிறுவயதில் மியர் பல மணி நேரம் பண்ணையில் உழைக்க வேண்டிய சூழ்நிலைகளுக்கிடையிலும், இயற்கை அவருக்கான ஓய்வுதரும் தலமாக இருந்தது.

கிடைத்த ஒவ்வொரு நேரத்திலும் அவர் காடுகளுக்குள் சென்று தாவரங்கள், விலங்குகள் ஆகியவற்றைக் கவனித்து ரசித்தார்.

ஒரு விபத்தில் தற்காலிகமாகப் பார்வையை இழந்த அனுபவம், மியரின் வாழ்க்கையை முற்றிலும் மாற்றியது. மீண்டும் பார்வை கிடைத்தால், இயற்கையை ஆராய்வதற்கே தனது வாழ்நாளை அர்ப்பணிப்பேன் என்று அவர் உறுதி எடுத்தார். பார்வை திரும்பியதும் அந்த உறுதியை அவர் நிறைவேற்றினார்.

அவர் ஆயிரம் மைல் தூரம் நடைபயணம் செய்து இயற்கையை ஆராய்ந்தார். அந்த அனுபவங்களை அவர் எழுதிய “A Thousand&Mile Walk to the Gulf” என்ற நூலில் பதிவு செய்தார். இந்தப் பயணம் அவரை கலிஃபோர்னியாவின் அழகிய Yosemite Valley பள்ளத்தாக்கிற்கு அழைத்துச் சென்றது.

யோசமடப் பள்ளத்தாக்கின் மலைகள், அருவிகள், காடுகள் ஆகியவற்றைக் கண்டபோது, மியர், ஓர் ஆன்மீக அனுபவத்தை உணர்ந்தார். “இயற்கை ஒரு கோவில்” என்று அவர் நம்பினார்.

மியர் தனது அனுபவங்களைக் கட்டுரைகள் மற்றும் நூல்களாக எழுதி மக்களிடம் கொண்டு சென்றார். அவரது எழுத்துக்கள் வாசகர்களை இயற்கையின் மடியில் கொண்டு சேர்த்தன. மரங்கள், ஆறுகள், மலைகள் அனைத்தும் ஒன்றோடொன்று இணைந்தவை என்ற உண்மையை அவர் வலியுறுத்தினார்.

19-ஆம் நூற்றாண்டின் இறுதியில், காடுகள் வெட்டப்பட்டு இயற்கை அழிக்கப்பட்டுக் கொண்டிருந்தது. இதைத் தடுக்க மியர் உறுதியுடன் போராடினார். அவரது முயற்சியால் 1890-ஆம் ஆண்டு Yosemite National Park உருவாக்கப்பட்டது.

மேலும் Sequoia National Park மற்றும் Mount Rainier National Park போன்ற பல தேசியப் பூங்காக்கள் உருவாக அவரின் எண்ணங்களும், எழுத்துக்களும் ஊக்கமளித்தன. மியருக்கு, இயற்கையைப் பாதுகாப்பது இயற்கையின் அழகைக் காக்கும் செயல் மட்டுமல்ல; அதை, அவர் மனித ஆன்மாவைப் பாதுகாக்கும் புனிதமான கடமையாகக் கருதினார்.

1903-ஆம் ஆண்டு அமெரிக்க அதிபர் Theodore Roosevelt, மியரை, யோசமைட் பகுதியில் சந்தித்தார். இருவரும் மூன்று நாட்கள் இயற்கையின் மத்தியில் தங்கிப் பேசினர். இந்தச் சந்திப்பு ரூஸ்வெல்டின் மனதில் ஆழமான தாக்கத்தை ஏற்படுத்தியது. அதன் பின்னர் அவர் பல தேசியப் பூங்காக்கள், காடுகள் மற்றும் வனவிலங்கு சரணாலயங்களை உருவாக்கினார்.

1892-ஆம் ஆண்டு, மியர் "Sierra Club" என்ற அமைப்பைத் தொடங்கினார். இது உலகின் மிக முக்கியமான சுற்றுச்சூழல் அமைப்புகளில் ஒன்றாக வளர்ந்தது. இந்த அமைப்பு இன்று வரை இயற்கையைப் பாதுகாக்கும் பணியில் தொடர்ந்து செயல்பட்டு வருகிறது.

1914-ஆம் ஆண்டு மியர் உயிரிழந்தாலும், அவரது எண்ணங்கள் இன்னும் உயிருடன் உள்ளன. தேசியப் பூங்காக்கள், காடுகள், இயற்கையை நேசிக்கும் மனிதர்கள் அனைத்திலும் அவரது தாக்கம் காணப்படுகிறது.

மியர் மனித குலத்திற்குக் கற்றுத்தந்த மிக முக்கியமான உண்மை என்னவென்றால்:

இயற்கையைப் பாதுகாப்பது ஒரு பொறுப்பு மட்டுமல்ல; அது ஒரு பக்தியுடன் கூடிய ஒரு புனிதமான செயல்.

ஜான் மியர் வாழ்க்கை, மனிதன் இயற்கையுடன் இணைந்து வாழ வேண்டும் என்பதை உணர்த்தும் ஒளிக்கதிராகும். இன்றைய உலகில் சுற்றுச்சூழல் சவால்கள் அதிகரித்து வரும் நிலையில், அவரது சிந்தனைகள் மிகவும் அவசியமானவை.



திரு. வ.சுந்தரராஜ் இ.வ.ப (ஓ)
முன்னாள் துணை வனப்பாதுகாவலர்

CONDOLENCE

TASPEF pays rich tributes to our distinguished former Joint Secretary Thiru M.K. Karunanidhi formerly Divisional Forest Officer, He took active part in TASPEF in initial years and made his presence felt. He also participated in all our AGM till his health permitted him and made valuable contribution to the management of our Association. TASPEF grieves his death and stands with his family. We thank Thiru V.Sambasivam IFS(Retd) for writing a moving tribute on him as his family friend. Let us pray for the eternal bliss of a Veteran Forester's soul.

நெய்தல் கொண்ட கண்டல் கண்டிரோ ?!...

கண்டல் நிறைந்த காடுகளை
கடற்கரையோரம் கண்டிரோ ?!
கானுயிர் கவர்ந்திழுக்கும் - கடலின்
கண்கவர் காவியம் காண்பீரே!!



உள்ளம் உருகும் காண்கையிலே!
உவர் படிமங்கள் இலைகளின்மேலே!!
சதுப்பு நிலங்களின் காடுகளே!
சாய்ந்தாடும் திருவாரூர் தேர்போலே!!



ஆறுகளின் முகத் துவாரங்களே!
பேரலைகளை தாங்கும் அச்சாரங்களே!!
சிவந்த நீரின் சிறுகாயல்களே - கடல்
சீற்றத்தை குறைக்கும் பசுஞ்சிகரங்களே!!



முளைத்த விதைகள் மரக்கிளைகளிலே!
முணுமுணுக்கும் வண்டுகள் பூக்களிலே!!
சுவாசவேர்களின் சுவாரஸ்யங்களை
சுமந்து நிற்கும் பசுமைச்சுரங்களே!!



பறவைகளின் பல வசனங்களை
பார் முழுதினிலிருந்தும் பெறுவீர்களே!
புயலையும் கொல்லும் புதல்வர்களே - நீங்கள்
புண்ணிய தேசத்துப் புதினங்களே!!



நிலவு ஒளியின் வெளிச்சத்திலே!
நீரினுள் இருக்கும் இறால்களை
நீந்திப் பிடிக்கும் மீனவர்களை
நித்தமும் காக்கும் நீலக்கடல் கானகமே!!



மக்களைக் காக்கும் மழைக்காடுகளே!
மாடுகளால் அழிக்கப்படுகிறீர்களே!!
சிறுபான்மைகளின் செல்வங்களே!
சிலப்பதிகாரத்தின் தொன்மங்களே!!



பல்கிப் பெருகும் பல்லுயிர்களுக்கும்
பதமான சூழலை பகிர்ந்தளித்திருப்பது!
பாரதத்தாயின் பன்முக தன்மைக்கு - தாங்கள்
பெருமை சேர்த்திடும் பேராண்மையன்றோ!!



என்றென்றும் பசுமை குடியிருக்கும்
எங்கள் நெய்தல் தேசத்துக் கோவில்களே!
உங்களைப் போற்றித் துதிப்பேனே -
உம்சிறப்பினை உலகம் வியக்க
உரைப்பேனே! உரைப்பேனே!!



[கண்டலின் மீது கொண்ட காதலால் எழுந்தது]

ஆ. பெரியசாமி இ.வ.ப.,
தலைமை வனப்பாதுகாவலர்,
தமிழ்நாடு வனத்துறை,

